

HW 4.2.1.c: Solving Trigonometric Equations (using algebra)

Find all solutions on the interval $0 \leq \theta < 2\pi$.

1. $2 \sin(\theta) = -1$ 2. $2 \sin(\theta) = \sqrt{3}$ 3. $2 \cos(\theta) = 1$ 4. $2 \cos(\theta) = -\sqrt{2}$

5. $2 \sin\left(\frac{\pi}{4}x\right) = 1$ 6. $2 \sin\left(\frac{\pi}{3}x\right) = \sqrt{2}$ 7. $2 \cos(2t) = -\sqrt{3}$ 8. $2 \cos(3t) = -1$

9. $\sin^2 x = \frac{1}{4}$ 10. $\cos^2 \theta = \frac{1}{2}$ 11. $\sec^2 x = 4$ 12. $3 \csc^2 t = 4$

13. $\csc(2x) - 2 = 0$

14. $\sec(2\theta) = 2$

15. $2 \sin(x) \cos(x) = \cos(x)$

16. $\sqrt{2} \sin(t) = 2 \cos(t) \sin(t)$

17. $\sec(x) \sin(x) - 2 \sin(x) = 0$

18. $\tan(x) \sin(x) - \sin(x) = 0$

19. $\tan^3(x) = 3 \tan(x)$

20. $\cos^3(t) = \cos(t)$

21. $\tan^3(x) = \tan(x)$

22. $\tan^5(x) - 9 \tan(x) = 0$

23. $2 \sin^2 w + 3 \sin w + 1 = 0$

24. $2 \cos^2 x + 3 \cos(x) + 1 = 0$

25. $2 \cos^2 t + \cos t = 1$

26. $2 \sin^2(\theta) + 3 \cos(\theta) - 3 = 0$

27. $2 \cos^2(\theta) = 2 + \sin(\theta)$

28. $4 \sin(x) \cos(x) + 2 \sin(x) - 2 \cos(x) - 1 = 0$

29. $2 \sin(x) \cos(x) - \sin(x) + 2 \cos(x) - 1 = 0$

Selected Answers: (Teachers feel free to add more solutions)

1. On the interval $0 \leq \theta < 2\pi$, $\theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$. Without this interval, $\theta = \frac{7\pi}{6} \pm 2n\pi$ or $\theta = \frac{11\pi}{6} \pm 2n\pi$

3. On the interval $0 \leq \theta < 2\pi$, $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$. Without this interval, $\theta = \frac{\pi}{3} \pm 2n\pi$ or $\theta = \frac{5\pi}{3} \pm 2n\pi$

5. On the interval $0 \leq \theta < 2\pi$, $x = \frac{2}{3}$ or $\frac{10}{3}$. Without this interval, $\theta = \frac{2}{3} \pm 8n$ or $\theta = \frac{10}{3} \pm 8n$

7. On the interval $0 \leq \theta < 2\pi$, $t = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$.

9. On the interval $0 \leq \theta < 2\pi$, this occurs at $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

11. On the interval $0 \leq \theta < 2\pi$, this occurs at $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

13. $x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}$

15. On the interval $0 \leq \theta < 2\pi$, $\cos x = 0$ at $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$, and $\sin x = \frac{1}{2}$ at $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$.
 $\sin x = 0$ $\sec x = 2$

17. $x = 0, \pi$ $\cos x = \frac{1}{2}$
 $x = \frac{\pi}{3}, \frac{5\pi}{3}$

19. On the interval $0 \leq \theta < 2\pi$, $\tan x = 0$ at $x = 0$ and $x = \pi$. Similarly, $\tan x = \sqrt{3}$ at $x = \frac{\pi}{3}$ and $x = \frac{4\pi}{3}$, and $\tan x = -\sqrt{3}$ at $x = \frac{2\pi}{3}$ and $x = \frac{5\pi}{3}$.

21. On the interval $0 \leq \theta < 2\pi$, $\tan x = 0$ at $x = 0$ and $x = \pi$. Similarly, $\tan x = 1$ at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$. Finally, $\tan x = -1$ for $x = \frac{3\pi}{4}$ and $x = \frac{7\pi}{4}$.

23. On the interval $0 \leq \theta < 2\pi$, $\sin w = -\frac{1}{2}$ at $w = \frac{7\pi}{6}$ or $w = \frac{11\pi}{6}$. Similarly, $\sin w = -1$ at $w = \frac{3\pi}{2}$

25. On the interval $0 \leq \theta < 2\pi$, $\cos t = \frac{1}{2}$ at $t = \frac{\pi}{3}$ or $t = \frac{5\pi}{3}$. Similarly, $\cos t = -1$ at $t = \pi$.
 $\sin \theta = 0$ $2 \sin \theta + 1 = 0$

27. $\theta = 0, \pi$ $\sin \theta = -\frac{1}{2}$
 $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$

29. Either $2\cos(x) - 1 = 0$ or $\sin(x) + 1 = 0$. This leads to $x = \frac{\pi}{3}$, $x = \frac{5\pi}{3}$ (for $\cos x = \frac{1}{2}$) and $x = \frac{3\pi}{2}$, (for $\sin x = -1$).